

SVKM
D. J. Sanghvi College of Engineering
Programme: B.Tech in Computer Engineering
Year: III/Semester VI (Exam Year: 2024-2025)

Subject: Information Security

Date: 24 May 2025

Time: 09:00 am - 11:00 am (02:00 Hrs.)

Max Marks: 65

FINAL EXAMINATION(2024-2025)

Instructions:

1. This question paper contains 3 pages
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Draw Neatly labelled Diagrams wherever applicable.
5. Assume Suitable data wherever required.

- | | | |
|------------------------------|---|----|
| 1. | | 20 |
| A. .. | | 10 |
| | i. Explain the encryption process of the Double Columnar Transposition Cipher with clear steps. Using the plaintext "informationsecurity" and the key "PLANT", demonstrate how the encryption is performed. | 10 |
| <p>----- OR -----</p> | | |
| | ii. Explain the steps involved to convert the plaintext in to cipher text using Playfair cipher. Convert the following Plaintext in to cipher text using Playfair cipher:
Key = THIRDCO , Plaintext = COMPUTER | 10 |
| B. | Describe the working of the AES encryption algorithm with operations performed in each round. | 10 |
| 2. | | 15 |
| A. .. | | 7 |
| | i. Give the steps in RSA algorithm. Perform encryption and decryption using the RSA | 7 |

algorithm for the following data: $p=5$; $q=11$; $e=3$; $M=9$

----- OR -----

ii. Mention the steps involved in the Diffie-Hellman Key Exchange Algorithm. In a Diffie-Hellman key exchange scenario, Alice and Bob agree on a prime number $n=23$ and a base $g=5$. 7

1. Alice selects a private key $x=6$. Compute her public key.

2. Bob selects a private key $y=15$. Compute his public key.

3. Compute the final shared secret key for both Alice and Bob.

B. Draw and explain the internal working of the Feistel Network that forms the backbone of the DES algorithm. For the given data, predict the output after Round 2 in the Feistel network. 8

■ Plaintext: 0011 0101

■ Operation inside the function block: OR

■ Key1 = 0110

■ Key2 = 1001

3. 15

A. What is a buffer overflow vulnerability in computer systems? Explain how it occurs with the help of a simple code example. 7

B. .. 8

i. What is MD5, and how does it transform input data into a fixed-size hash value? 8

----- OR -----

ii. What is HMAC, and how is it different from CMAC? 8

4. 15

7

A. What is Denial of Service (DoS) Attack? Explain Distributed Denial of Service (DDoS) attack in detail and outline the strategies to employ defense against DDoS attack.

B. ..

8

1.

8

What is a firewall? List common types of firewalls and briefly explain any two.

----- **OR** -----

2. What is DNS spoofing in information security? Explain how DNS spoofing attacks are carried out and describe strategies to mitigate such attacks.

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